Sahaj Bhadja

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Summary

Software Engineer with 2+ years of experience building **React/TypeScript applications**, **Node.js services**, and responsive **high-throughput workflows** using REST APIs, real-time data, and performance-focused engineering.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Tripled throughput across 1K+ AI jobs by restructuring **concurrent backend orchestration** with explicit job states, failure isolation, recoverable processing, and observable workflows under production load reliably daily.
- Reduced dashboard latency from 1.2s to 350ms by optimizing **indexed SQL queries** and multi-tenant data models supporting 500+ users with normalized schemas, row-level isolation, and efficient access patterns.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Reduced feedback turnaround 60% by delivering a **React and Flask platform** for 150+ instructors and 10K+ daily grading requests, integrating responsive interfaces, REST APIs, AWS RDS, and CI/CD.
- Processed 1K+ files per batch using **Python multithreading**, validating heterogeneous inputs while isolating failures and preserving responsive workflows during concurrent model execution and production traffic reliably at scale.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while troubleshooting integrations across Windows, Linux, and macOS production environments at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins and GitHub**, adding failure reports and cross-platform validation across Linux, Windows, and macOS with repeatable nightly workflows at scale.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Synchronized 2M+ records through an eight-service **real-time microservice platform** using React, TypeScript, Node.js, Express.js, WebSockets, REST APIs, and Azure SQL across distributed customer-facing business workflows at scale reliably.
- Increased lead conversions 15% by delivering **responsive CRM workflows** for customers, billing, resource allocation, notifications, and role-based access with automated testing, monitoring, and Docker/Jenkins deployments across eight service domains.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built **cross-platform booking workflows** spanning mobile and web, managing time slots, pricing, coupons, payments, refunds, disputes, and real-time Firebase synchronization across 50K+ customer booking records with consistent cross-platform experiences.
- Delivered sub-200ms **Flask booking APIs** with payment webhooks, Firebase transactions, and status synchronization, reducing confirmation time 40% and manual refund handling approximately 70% across transactional workflows reliably.

Skills

Languages: JavaScript, TypeScript, React, Node.js, Python, Java, SQL, HTML, CSS
Frontend & Design: Responsive Design, Next.js, TailwindCSS, Shadcn, MVC, Modular Architecture, WebSockets, Cross-Platform Development
Backend & APIs: Express.js, Flask, FastAPI, REST APIs, Microservices, OOP, API Design, PostgreSQL, Azure SQL, MongoDB
Cloud & Delivery: AWS, Azure, GCP, Docker, Kubernetes, CI/CD, GitHub Actions, Jenkins, Git
Engineering: Performance Tuning, Caching, Multithreading, Automated Testing, Unit Testing, Integration Testing, PyTest, Production Troubleshooting, System Design

Projects

High-Throughput Travel Search Gateway - Design Concept

- Designed a **Node.js and GraphQL travel-search gateway** concept unifying hotel, flight, rental-car, and package queries with schema validation, caching, pagination, error isolation, and observable API workflows reliably.

Multi-Platform Travel Results Experience - Design Concept

- Designed a **React and TypeScript travel-results interface** concept using modular components, responsive layouts, lazy loading, route-level code splitting, cached filters, and render-performance instrumentation across desktop, mobile, and tablet.

FinNexus | Live

- Built a responsive **TypeScript transaction platform** supporting 50K+ users and 1M+ transactions using Next.js, Kubernetes, MongoDB, WebSockets, secure APIs, real-time synchronization, advanced filtering, and CI/CD workflows at scale.